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课堂笔记整理4.7

7.2节

例7-1

$R_{eq}=20\|5=4\Omega$，$\tau=R_{eq}C=4\times0.1=0.4s$，

$v=v(0)e^{-\frac{t}{\tau}}=15e^{-\frac{t}{0.4}}V$，则 $v_c=v=15e^{-2.5t}V$

$v_x=\frac{12}{12+8}v=0.6\times(15e^{-2.5t})=9e^{-2.5t}V$

$i_x=\frac{v_x}{12}=0.75e^{-2.5t}V$

练7-1

$R_{(6\|12)}=6\|12=4\Omega$，$R_{eq}=8+4=12\Omega$

$\tau=R_{eq}C=12\times\frac{1}{3}=4s$，$v_c=v(0)e^{-\frac{t}{\tau}}=60e^{-0.25t}$

$v_x=\frac{4}{4+8}v_c=\frac{1}{3}\times60\times e^{-\frac{t}{4}}=20e^{-0.25t}$，$i_c=-\frac{v_c}{R_{eq}}=-5e^{-0.25t}$

例7-2

$V_c(t)=\frac{9}{9+3}20=15V$，$R_{eq}=1+9=10\Omega$，$\tau=R_{eq}C=10\times20\times10^{-3}=0.2s$

$v(t)=v_c(0)e^{-\frac{t}{\tau}}=15e^{-\frac{t}{0.2}}V=15e^{-5t}V$，$W_c(0)=\frac{1}{2}Cv_c^2(0)=\frac{1}{2}\times20\times10^{-3}\times15^2=2.25J$

练7-2

$R_{eq}=4\|12=3\Omega$，$V_c(0)=\frac{3}{9}\times24=8V$，$\tau=R_{eq}C=3\times\frac{1}{6}=0.5s$

$v=8e^{-\frac{t}{0.5}}=8e^{-2t}V$，$W_0=\frac{1}{2}Cv^2=\frac{1}{2}\times\frac{1}{6}\times8^2=5.333J$

7.3节

例7-3

可列得如下方程：

$$\begin{cases}2(i_1-i_2)+\frac{1}{2}\frac{di_1}{dt}=0\\-3i+4i_2+2(i_2-i_1)=0\\i=i_1\end{cases}$$

整理后可得 $\frac{di_1}{dt}+\frac{2}{3}i_1=0$，即 $\frac{di_1}{i_1}=-\frac{2}{3}dt$

对等式两边积分得：$\ln i\Big|_{i(0)}^{i(t)}=-\frac{2}{3}t\Big|_0^t$，即 $\ln\frac{i(t)}{i(0)}=-\frac{2}{3}t$.

即 $i(t)=i(0)e^{-\frac{2}{3}t}=10e^{-\frac{2}{3}t}A$，

故 $v=L\frac{di}{dt}=0.5\times10\times(-\frac{2}{3})e^{-\frac{2}{3}t}=-\frac{10}{3}e^{-\frac{2}{3}t}V$

则 $i_x(t)=\frac{v}{2}=-1.667e^{-\frac{2}{3}t}A$

练7-3

$-1+V_x+6\times I_2=0$，$-2V_x+2(I_2-I_1)+6\times I_2=0$，$V_x=1\times I_1$，

$I_1=\frac{1}{4}$，$I_2=\frac{1}{8}$，$R_{eq}=\frac{1}{I_1}=4\Omega$，$\tau=\frac{L}{R}=\frac{2}{4}=0.5s$

则 $i(t)=7e^{-2t}A$，$t>0$，$V_x=-I\times1=-7e^{-2t}A$，$t>0$.

例7-4

$4\|12=3\Omega$，$I_{总}=\frac{40}{2+3}=8A$，$I_0=\frac{12}{12+4}\times8=6A$，

$R_{eq}=16\|16=8\Omega$，$\tau=\frac{L}{R}=\frac{2}{8}=\frac{1}{4}$，$i(t)=I_0e^{-\frac{t}{\tau}}=6e^{-4t}$

练7-4

$t<0$时，电感短路，即5Ω电阻两端视作开路，$8 \parallel 24 = 6\Omega$

$I_0 = \frac{6}{6+12} \times 6 = 2A$，$R_{eq} = (8+12) \parallel 5 = 4\Omega$，$\tau = \frac{L}{R} = \frac{2}{4} = 0.5$

$t>0$时，

$i(t) = I_0 e^{-\frac{t}{\tau}} = 2e^{-2t}A$

例7-5

$t<0$时，$i_0 = 0$，$i(t) = \frac{10}{2+3} = 2A$，$t<0$，$V_0(t) = 3i(t) = 6V$，$t<0$

$t>0$时，$R_{Th} = 3 \parallel 6 = 2\Omega$，$\tau = \frac{L}{R_{Th}} = 1s$，$i(t) = i(0)e^{-\frac{t}{\tau}} = 2e^{-t}A$，$t>0$

$V_0(t) = -V_L = -L\frac{di}{dt} = -2(-2e^{-t}) = 4e^{-t}V$，$t>0$

$i_0(t) = \frac{V_L}{6} = -\frac{2}{3}e^{-t}A$，$t>0$

练7-5

$t<0$时，

$i = \frac{4}{6} \times 24 = 16A$，$V_0 = 16 \times 2 = 32V$，$i_0 = \frac{2}{6} \times 24 = 8A$

$t \geq 0$时，

$R_{eq} = (4+2) \parallel 3 = 2\Omega$，$\tau = \frac{L}{R} = \frac{1}{2}$，$i(t) = 16e^{-2t}A$

$V_L(t) = L \times \frac{di}{dt} = 1 \times 16 \times (-2)e^{-2t} = -32e^{-2t}V$

$i_3(t) = -\frac{32}{3} \times e^{-2t}A$，$i_0(t) = -(i_3(t) + i(t)) = \frac{32}{3}e^{-2t} - 16e^{-2t} = -5.333e^{-2t}A$，

$V_0(t) = -2 \times i_0(t) = \frac{32}{3} \times e^{-2t} = 10.667e^{-2t}V$

则 $i = \begin{cases} 16A, & t<0 \\ 16e^{-2t}, & t \geq 0 \end{cases}$ $i_0 = \begin{cases} 8A, & t<0 \\ -5.333e^{-2t}A, & t>0 \end{cases}$，$V_0 = \begin{cases} 32V, & t<0 \\ 10.667e^{-2t}V, & t>0 \end{cases}$